Leadership & Consulting in Data Science

Week 4



Team Presentation – McDonald's All-Day Breakfast



Create the ***draft*** set of analyses to consider sharing with the client

- Consider importance of insights and relevance to problem
- Determine order…keeping storytelling in mind

Review the EDA generated by each of your team members

Create a list of questions that would help clarify your understanding further

Reconvene with the class and compare your key EDA findings with the class

- Create a short (1-3 visuals) from your EDA that tells a story
- A team member with the visuals should join the class Zoom session
- Each team will share their story (they do NOT have to be slides)
- Class should be prepared to provide feedback

Week 4 Agenda

1



Project Management

- Project Definition
- Anatomy of a Proposal
- Agile and Scrum Methodology
- Scheduling and Budgets

Projects need to be well defined and agreed upon



Problem Statement

Criteria for acceptance

- Deliverables
- Quality / Performance Metrics



Time Constraint

Resources

- Sponsor / Client – committed resources, availability, and responsiveness
- Team / Vendor – **qualifications**, committed resources, availability, and responsiveness
- Other (e.g., $s, supplies)

"Our Understanding" and Objectives

Consultant understands Client's data and analytics needs



Our understanding of the current situation

- Client is a wholly owned subsidiary of Parent and provides consumer financial products and services to Brand A, Brand B and Brand C dealers and their customers
- Client has a key initiative to develop its ability to transform data into business insights and outcomes and wants to understand what is required to advance its analytics capabilities at a department level
- Although some of Client's business groups are exploring analytics and insights, most teams lack the time or expertise to address this analytics initiative and therefore Client has created a cross-functional team to lead this initiative
- Client is looking for objective analytics expertise to guide an evaluation (e.g., people, skills, process, technology, culture), to identify best-in-class capabilities, design a target state with Client's capabilities in mind, and provide a roadmap and recommendations to build Client's analytics capabilities for a single VCI department with an eye on repeating this for other departments



Consultant Engagement

- Consultant will work with Client on a 10 week, 4 step project to build Client's Analytics Roadmap and Recommendations for a specific department
- Consultant will conduct interviews with key stakeholders, review relevant Client documentation; these will be summarized, annotated based on best practices and Consultant expertise in "Initial Findings" and "Maturity Assessment"; a workshop will validate and finalize these deliverables
- Consultant will work with Client to understand the department's business needs and goals; based on the findings, maturity assessment, and goals, Consultant will propose the best practices that have to be adopted. Then Consultant will further modify them to fit Client's specific circumstances and thus create the analytics "Target State" and identify misses in the "Gap Analysis"; a workshop will validate and finalize these deliverables
- Consultant will then work with Client to create a prioritized and sequenced "Roadmap" and detail out "Recommendations" in order to build out the Operating and Technology Models to enhance the Client department's analytics desired state and infrastructure and provide this information in a final read out

Scope

This project will focus on Analytics for one department evaluated over Identified Building Blocks

Use Case (expresses the desired and intended purpose of use of the analytics)

- Client identified department-based analytical use cases within the context of the organization's culture

Topic (expresses the involved technical disciplines)

- Analytics

Building Blocks (the relevant focus based on the Gartner Seven Building Blocks framework)

- **In Scope**: Vision, Strategy, Information Governance, Organization and Roles, Information Life Cycle
- **Out of Scope**: Metrics, Enabling Infrastructure

Process

- This Business Analytics Strategy, Target State and Roadmap will appropriately leverage Consultant's Enterprise Information Management (EIM) Framework and include Mode 2 Analytics Workbench and Data Science Lab feeding into a more traditional Mode 1 Information Portal
  - Provide full design for the strategy, vision, metrics, and enabling infrastructure
  - Provide target state implications on information governance, organization & roles, and information life cycle
  - Roadmap will detail out requirements to build out and address implications of organizational change and Mode 1 vs. Mode 2 operation
- This project will incorporate learnings from, and not duplicate efforts of, the preceding Consultant EIM and Security engagements

Approach w/Deliverables

Consultant recommends a five (5) step approach to deliver the Roadmap and Initiate Implementation

	1 Project Kickoff and Management	2 Requirements Development	3 Architecture Design	4 Roadmap Development	5 Vendor Selection
Objectives	• Control Engagement Execution	• Finalize, normalize, de-conflict and rank order requirements • Decompose requirements into tool and architectural components	• Formulate the required technical architecture	• Establish the implementation roadmap for deploying the architecture	• Devise evaluation factors to assess the requirements • Conduct evaluation against the factors
Activities	• Conduct Kickoff • Conduct ongoing Status Meetings	• Review documentation • Conduct stakeholder interviews • Identify business strategies, drivers, and pain points • Review and finalize the requirements • Decompose requirements into tool and architectural components • Normalize and de-conflict the requirements • Rank order the requirements	• Map requirements to metadata types and domains • Map requirements to metadata tool classes • Examine existing metadata sources and platforms • Design logical and physical metadata architecture	• Review and analyze existing architecture components • Compare and identify gaps • Establish actionable recommendations for addressing gaps • Time-phase and sequence recommendations into project-level efforts	• Identify solution space and down-select • Design of PoC • Formalize evaluation model for factors and weights • Vendor alternatives evaluation (RFIs, RFPs) • Score and analyze vendors for down select
Deliverables	• Kickoff Materials • Project Plan • Status Reports	• Design Requirements Matrix • Draft PoC candidates	• Logical Architecture Design • Physical Architecture Design	• Roadmap	• PoC objectives and evaluation criteria • Vendor Evaluations and Scoring for down select
		• Draft PoC candidates	• Narrow PoC requirements and candidates		• Design PoC

Approach Details with Deliverables (example dates provided)

Consultant drives the ten (10) week engagement with three key workshops and weekly status meetings

Week 1: 9/4 – 9/8	Week 2: 9/11 – 9/15	Week 3: 9/18 – 9/22	Week 4: 9/25 – 9/29	Week 5: 10/2 – 10/6	Week 6: 10/9 – 10/13	Week 7: 10/16 – 10/20	Week 8: 10/23 – 10/27	Week 9: 10/30 – 11/3	Week 10: 11/6 – 11/10

Kickoff — Steering Committee — Final Readout

1. Kickoff

1. Project management

2. Enterprise Assessment and Analysis

Workshop 1: Findings and Assessment

3. Vision and Strategy Development

Workshop 2: Vision & Strategy

Workshop 3: Target State

4. Desired Target State and Gap Analysis

5. Roadmap and Recommendations

Key Deliverables / Activities

- Kickoff Materials
- Project Plan
- Status Reports
- Analytics Findings
- Analytics Maturity Assessment
- Analytics Vision and Strategy
- Analytics Strategic Plan
- Analytics Desired Target State
- Analytics Gap Analysis
- Analytics Roadmap
- Analytics Recommendations

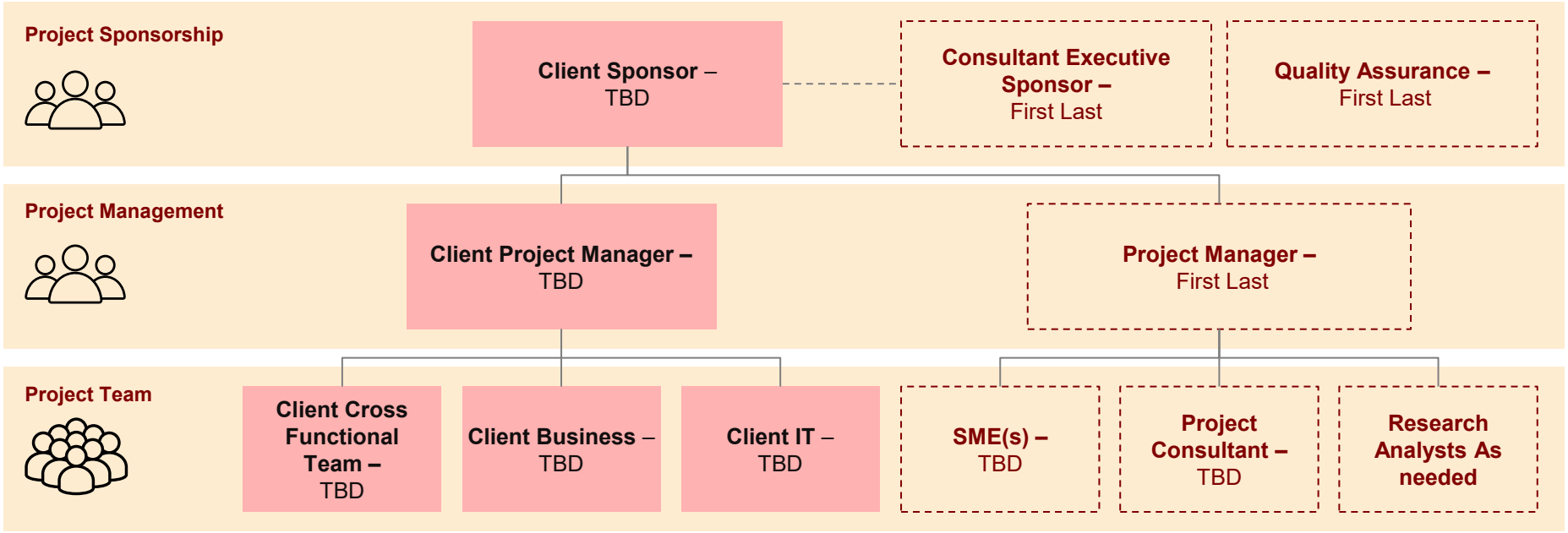
Meeting Legend ▲ Steering Committee ● Status ◆ Workshop

Note: This schedule is dependent on the assumptions included in this Statement of Work

Project Team

Organization Chart

Consultant has created an organization structure for this project that ensures high-level sponsorship and quality assurance, strong day-to-day project management, a focused team of project consultants, and deep subject matter expertise



Client Team Commitments

Project Team Member	**Step 0** Kickoff and Project Management	**Step 1** Enterprise Assessment and Analysis	**Step 2** Desired Target State and Gap Analysis	**Step 3** Roadmap and Recommendations
Duration	kickoff meeting + ongoing	4 weeks	4 weeks	2 weeks
Sponsor	8 hours	4 hours	4 hours	2 hours
Project Manager	24 hours	16 hours / week	8 hours / week	8 hours / week
Core Participants	8 hours / week	16 hours / week	8 hours / week	8 hours / week
Business Executives and Leaders		1 hour interview 4 hours – workshop (subgroup)	4 hours – workshop (subgroup)	2 hours – final readout (subgroup)
Business Managers and Support Personnel		1 hour interview 4 hours – workshop (subgroup)	4 hours – workshop (subgroup)	2 hours – final readout (subgroup)
IT and Technical Representatives		1 hour interview 4 hours – workshop (subgroup)	4 hours – workshop (subgroup)	2 hours – final readout (subgroup)
Business SMEs			2 hours / week	2 hours / week
Technical SMEs			2 hours / week	2 hours / week

Managing Project Risk

Identify risks early, log them in a risk register with owners and mitigations, and revisit the register at every status meeting



Traditional Project Risks

- Scope creep
- Resource availability
- Schedule and budget overruns
- Vendor and dependency delays



AI and Data Science Risks

- Missing, dirty, or delayed data
- Model misses accuracy target
- Bias, privacy, compliance
- Model drift after deployment
- Uncertain experiment outcomes
- Key-person / scarce-skill risk
- Overtrust in model outputs

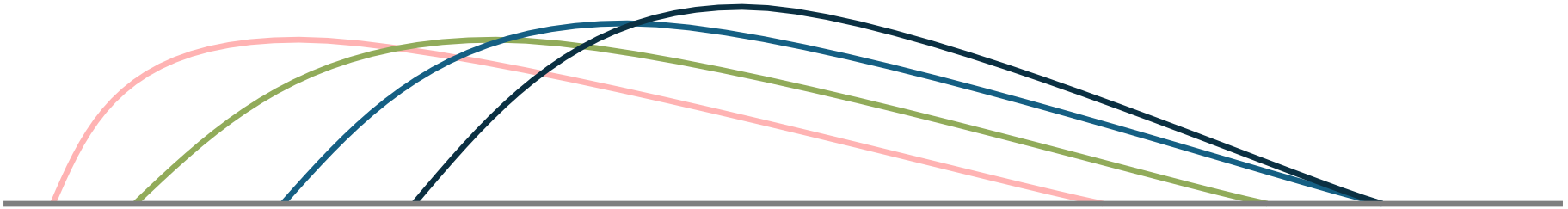
Software Development Lifecycle

Agile methods have mostly replaced the traditional sequential waterfall approach

Sequential Waterfall Approach

Requirements	Design	Implementation	Verification

Agile Overlapping Approach



Source: "The New New Product Development Game", Hirotaka Takeuchi and Ikujiro Nonaka, Harvard Business Review, January 1986

Agile Methodology

Agile is an iterative approach which uses cross-disciplinary teams to rapidly build software products and solutions



Benefits

- Faster time to market
- Increased customer satisfaction
- Values employees
- Minimizes rework



Characteristics

- Transparent
- Seamless communication
- Driven by trust and teamwork
- Minimal documentation
- Pair programming
- Rough estimates
- Fail fast

Agile Methods



01	02	03	04	05
Scrum	Kanban	Extreme Programming (XP)	Feature Driven Development (FDD)	Lean Software Development (LSD)

Agile Scrum Process

Hands-on system consisting of simple interlocking steps and components



Source: Adapted from Agile Software Development with Scrum by Ken Schwaber and Mike Beedle.

The Scrum Team



Typically 5-10 people

Cross-functional

- Business, Data Engineering, Data Science, QA, Developers, UI Designers, etc

Members should be full-time

- With some exceptions (e.g., System Admin, etc.)

Teams are self-organizing

- What to do if a team self-organizes someone off the team??
- Ideally, no titles but rarely a possibility

Membership can change only between sprints

Primary Roles



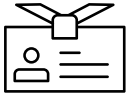
Scrum Master

- Servant-leader and coad for the Scrum Team
- Accountable for establishing and upholding Scrum practices
- Keeps the team focused on the sprint goal
- Removes impediments that block team progress



Product Owner

- Represents the clients/stakeholders
- Responsible for product features and requirements
- Typically filled by a Business domain expert
- Main job is to build out product backlog, prioritize it and answer questions



Team Member

- All other members of the agile team

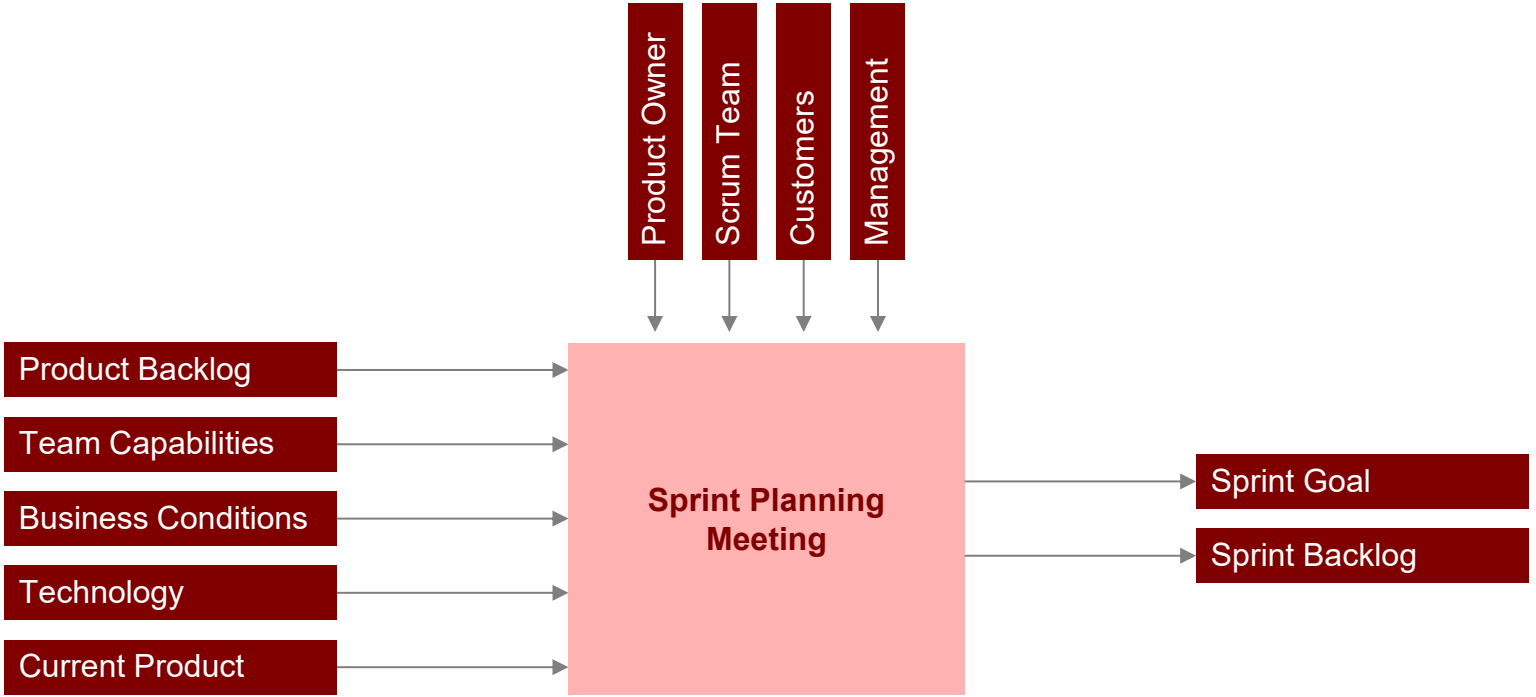
Business Requirements

	User Story	Epic
	• Smallest unit of deliverable value • Informal, general explanation of a software feature written from the perspective of the end user or customer	• A collection of related user stories
Example	• As a Risk manager (role) I would like the lending application to predict the chances of loan default (what) for a given individual based on several factors so that I can make an informed lending decision (why)	"Customer Risk Evaluation Epic"

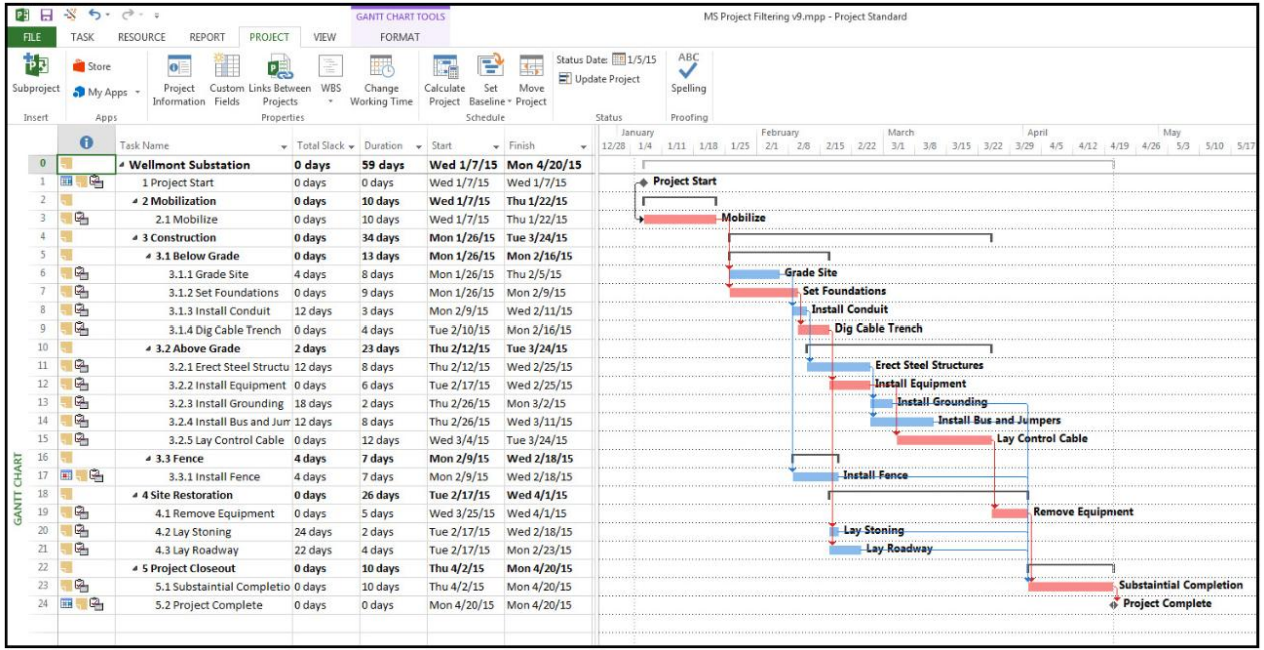
Scrum Calendar

Monday	Tuesday	Wednesday	Thursday	Friday
16	17	18 **Sprint Planning (Sprint 1)**	19 Daily standup	20 Daily standup
23 Daily standup	24 Daily standup	25 Daily standup **Backlog Refinement**	26 Daily standup	27 Daily standup
30 Daily standup	31 Daily standup	1 Daily standup **Backlog Refinement**	2 Daily standup	3 Daily standup
6 Daily standup	7 **Sprint Review (Sprint 1)** **Sprint Retrospective (Sprint 1)**	8 **Sprint Planning (Sprint 2)**	9 Daily standup	10 Daily standup

Sprint Planning



Gantt Chart – Sample MS Project Gantt Chart



	Task Name	Total Slack	Duration	Start	Finish
0	**Wellmont Substation**	**0 days**	**59 days**	**Wed 1/7/15**	**Mon 4/20/15**
1	1 Project Start	0 days	0 days	Wed 1/7/15	Wed 1/7/15
2	**2 Mobilization**	**0 days**	**10 days**	**Wed 1/7/15**	**Thu 1/22/15**
3	2.1 Mobilize	0 days	10 days	Wed 1/7/15	Thu 1/22/15
4	**3 Construction**	**0 days**	**34 days**	**Mon 1/26/15**	**Tue 3/24/15**
5	**3.1 Below Grade**	**0 days**	**13 days**	**Mon 1/26/15**	**Mon 2/16/15**
6	3.1.1 Grade Site	4 days	8 days	Mon 1/26/15	Thu 2/5/15
7	3.1.2 Set Foundations	0 days	9 days	Mon 1/26/15	Mon 2/9/15
8	3.1.3 Install Conduit	12 days	3 days	Mon 2/9/15	Wed 2/11/15
9	3.1.4 Dig Cable Trench	0 days	4 days	Tue 2/10/15	Mon 2/16/15
10	**3.2 Above Grade**	**2 days**	**23 days**	**Thu 2/12/15**	**Tue 3/24/15**
11	3.2.1 Erect Steel Structu	12 days	8 days	Thu 2/12/15	Wed 2/25/15
12	3.2.2 Install Equipment	0 days	6 days	Tue 2/17/15	Wed 2/25/15
13	3.2.3 Install Grounding	18 days	2 days	Thu 2/26/15	Mon 3/2/15
14	3.2.4 Install Bus and Jum	12 days	8 days	Thu 2/26/15	Wed 3/11/15
15	3.2.5 Lay Control Cable	0 days	12 days	Wed 3/4/15	Tue 3/24/15
16	**3.3 Fence**	**4 days**	**7 days**	**Mon 2/9/15**	**Wed 2/18/15**
17	3.3.1 Install Fence	4 days	7 days	Mon 2/9/15	Wed 2/18/15
18	**4 Site Restoration**	**0 days**	**26 days**	**Tue 2/17/15**	**Wed 4/1/15**
19	4.1 Remove Equipment	0 days	5 days	Wed 3/25/15	Wed 4/1/15
20	4.2 Lay Stoning	24 days	2 days	Tue 2/17/15	Wed 2/18/15
21	4.3 Lay Roadway	22 days	4 days	Tue 2/17/15	Mon 2/23/15
22	**5 Project Closeout**	**0 days**	**10 days**	**Thu 4/2/15**	**Mon 4/20/15**
23	5.1 Substaintial Completio	0 days	10 days	Thu 4/2/15	Mon 4/20/15
24	5.2 Project Complete	0 days	0 days	Mon 4/20/15	Mon 4/20/15

Critical Path and Dependencies

The critical path is the longest chain of dependent tasks – it sets the minimum possible project duration



Why It Matters

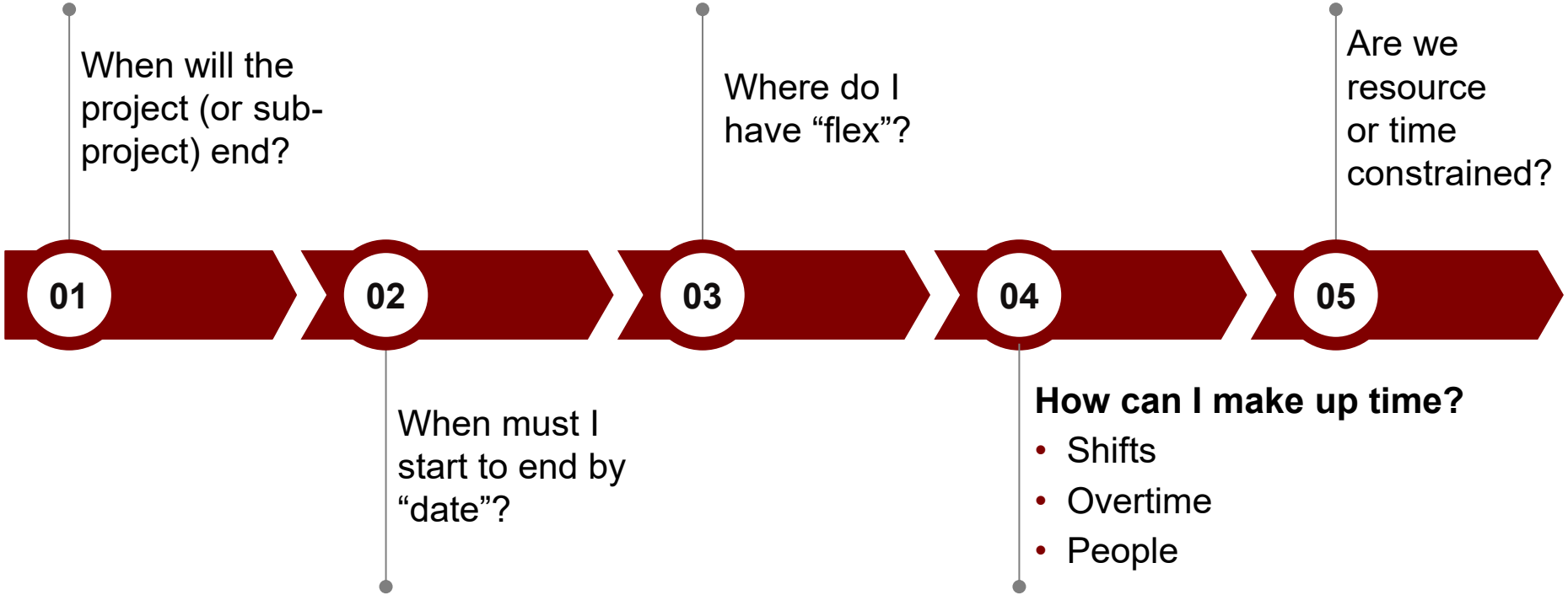
- A critical-path delay delays the project
- Slack shows where the schedule can flex
- Focuses attention on tasks that matter
- Answers “when will we finish?”



Key Concepts

- Dependency: must wait on another task
- Critical path: zero-slack chain
- Slack (float): allowable slip
- Milestone: zero-duration checkpoint
- Crashing: add resources
- Fast-tracking: overlap tasks
- Visualized on the Gantt chart

Project Arithmetic of Interest



End of Deck